

CLASS 03 · WEB DEVELOPMENT COURSE

Tables & Data Presentation

How HTML structures rows and columns from `<table>` basics to accessible, well-styled data grids.

`<table>``<tr><th><td>``thead/tbody/tfoot``colspan & rowspan``borders & spacing`

Duration: 60 minutes | Format: lecture + live coding

[Roadmap](#)

Today's Roadmap

- | | | |
|----|--|--------|
| 01 | Why tables exist & the core <code><table>/<tr>/<th>/<td></code> structure | 15 min |
| 02 | Semantic grouping: <code><thead></code> , <code><tbody></code> , <code><tfoot></code> , <code><caption></code> | 12 min |
| 03 | Merging cells: <code>colspan</code> & <code>rowspan</code> | 12 min |
| 04 | Columns as groups: <code><colgroup></code> / <code><col></code> | 4 min |
| 05 | Styling: borders, spacing, deprecated attributes | 9 min |
| 06 | Accessibility (scope/headers) + cheat sheet + exercise | 8 min |

Foundations

Why Do We Need Tables?

Tabular data is any data that naturally has rows and columns the same relationship repeats for every entry.

- Class timetables
- Price lists / pricing plans
- Spreadsheet exports, comparison charts
- Sports scores, financial reports

RULE OF THUMB

Use `<table>` only for real tabular data. Never use it for page layout that's CSS's job (flexbox/grid). Tables-for-layout was common in the 1990s–2000s and is now bad practice.

timetable.html — rendered

	Mon	Wed	Fri
9:00	HTML	CSS	JS
11:00	Lab	Lab	Project

Structure

The `<table>` Element

`<table>`

is the root container for every table. Everything else rows, headers, cells must live inside it.

Think of it as the outer box; on its own it renders nothing visible.

```
<table>  
  <!-- rows go here -->  
</table>
```

Structure

Rows & Data Cells: <tr> and <td>

<tr> = table row. It wraps one horizontal line of the table.

<td> = table data cell. A single cell of ordinary content, placed inside a <tr>.

Every <tr> should contain the same number of cells so columns line up.

```
<table>
  <tr>
    <td>Ali</td>
    <td>22</td>
  </tr>
  <tr>
    <td>Sara</td>
    <td>24</td>
  </tr>
</table>
```

Structure

Header Cells: <th>

<th> = table header cell. Labels what a row or column means.

- Rendered bold and centered by default (just default CSS — overridable)
- Carries real semantic meaning: screen readers announce it as a header
- <td> has none of that meaning — it's just content

BEST PRACTICE

Always use <th> for row/column labels instead of bolding a <td> with CSS — it's the difference between looking like a header and being one.

preview

Name	Score
Ali	88
Sara	91

[Live demo](#)

Putting It Together: tr + th + td

```
<table>
  <tr>
    <th>Product</th>
    <th>Price</th>
  </tr>
  <tr>
    <td>Notebook</td>
    <td>$3</td>
  </tr>
  <tr>
    <td>Pen</td>
    <td>$1</td>
  </tr>
</table>
```

preview

Product	Price
Notebook	\$3
Pen	\$1

Semantics

~3 min

Why Group Rows At All?

A plain table is just a flat stack of `<tr>` rows. As tables grow, we need to tell the browser (and assistive tech) which rows are headers, which are the actual data, and which are summary/total rows.

That's the job of three grouping elements: `<thead>`, `<tbody>`, `<tfoot>`.

`<thead>`

Header row(s) — column titles

`<tbody>`

The main data rows

`<tfoot>`

Totals / summary row(s)

Semantics

~2 min

<thead> — the Header Group

Wraps the row(s) that contain your <th> column labels. Usually just one row.

A table should have at most one <thead>.

```
<table>
  <thead>
    <tr>
      <th>Item</th>
      <th>Qty</th>
    </tr>
  </thead>
  ...
</table>
```

Semantics

<tbody> — the Body Group

Wraps the actual data rows. A table can have more than one <tbody> to split data into logical chunks.

If you omit <tbody>, browsers insert one invisibly — so it's always there whether you write it or not.

```
<tbody>
  <tr>
    <td>Pens</td>
    <td>12</td>
  </tr>
  <tr>
    <td>Books</td>
    <td>5</td>
  </tr>
</tbody>
```

Semantics

<tfoot> — the Footer Group

Wraps summary rows — totals, averages, grand sums. Classic use case: an invoice's total row.

GOOD TO KNOW

Even though <tfoot> is traditionally written after <tbody>, modern HTML lets it appear before <tbody> in the source — it still renders at the bottom.

```
<tfoot>
  <tr>
    <td>Total</td>
    <td>17</td>
  </tr>
</tfoot>
```

[Live demo](#)

thead + tbody + tfoot Together

```
<table>
  <thead>
    <tr><th>Item</th><th>Qty</th></tr>
  </thead>
  <tbody>
    <tr><td>Pens</td><td>12</td></tr>
    <tr><td>Books</td><td>5</td></tr>
  </tbody>
  <tfoot>
    <tr><td>Total</td><td>17</td></tr>
  </tfoot>
</table>
```

preview

Item	Qty
Pens	12
Books	5
Total	17

Often missed

<caption> — the Table's Title

<caption> gives the whole table a title/description. It must be the first child right after the opening <table> tag.

Screen readers announce it before reading the table — much better for accessibility than a floating <h3> above the table.

```
<table>
  <caption>
    Weekly Class Schedule
  </caption>
  <tr><th>Day</th><th>Topic</th></tr>
  ...
</table>
```

Merging cells

colspan — Merge Cells Horizontally

colspan="n" makes one cell stretch across n columns. Common for a heading that spans the whole table width.

```
<tr>
  <th colspan="3">Q1 Report</th>
</tr>
<tr>
  <th>Jan</th><th>Feb</th><th>Mar</th>
</tr>
```

preview

Q1 Report		
Jan	Feb	Mar
120	140	160

Merging cells

rowspan — Merge Cells Vertically

rowspan="n" makes one cell stretch down across n rows. Useful when a value repeats for several rows below it.

```
<tr>
  <td rowspan="2">Morning</td>
  <td>HTML</td>
</tr>
<tr>
  <td>CSS</td>
</tr>
```

WATCH OUT

When a cell spans multiple rows, every row below it needs one fewer `<td>` — that column's cell is already "filled" by the span.

preview

Slot	Class
Morning	HTML
	CSS

[Live demo](#)

Combining colspan & rowspan

```
<table>
  <tr>
    <th>Time</th>
    <th colspan="2">Monday</th>
  </tr>
  <tr>
    <td rowspan="2">9-11</td>
    <td>HTML</td>
    <td>Lab</td>
  </tr>
  <tr>
    <td>CSS</td>
    <td>Lab</td>
  </tr>
</table>
```

preview

Time	Monday	
9-11	HTML	Lab
	CSS	Lab

Often missed

<colgroup> & <col> — Styling Whole Columns

<colgroup> groups one or more **<col>** elements to style entire columns at once — without touching every <td>.

span="2" on <col> applies to two columns in one line.

Must appear right after <caption> (if present), before any row.

```
<table>
  <colgroup>
    <col style="background:#eee">
    <col span="2">
  </colgroup>
  <tr>
    <th>Name</th><th>A</th><th>B</th>
  </tr>
</table>
```

Styling

Borders: the Old Way vs. the CSS Way

LEGACY (AVOID)

`<table border="1">` — an HTML attribute. It still works, but it mixes presentation with structure. Deprecated in modern practice.

MODERN (USE THIS)

`table, th, td { border: 1px solid #333; }` — controlled entirely in CSS, easy to theme and reuse across a whole site.

Styling

border-collapse & border-spacing

By default, adjacent cell borders render as two separate lines. `border-collapse: collapse;` merges them into a single shared line — the clean, standard look.

`border-spacing` only matters when collapse is NOT used — it sets the gap between cells (replaces the old cellpadding attribute).

collapse

A	B
1	2

separate + spacing

A	B
1	2

```
table {  
  border-collapse: collapse;  
  /* OR, if separate: */  
  border-spacing: 6px;  
}
```

Styling

Cell Padding & Text Alignment

```
th, td {  
  padding: 8px 12px;  
  text-align: left;  
  vertical-align: middle;  
}  
th { text-align: center; }
```

- padding — breathing room inside each cell (replaces old cellpadding attribute)
- text-align — left / center / right
- vertical-align — top / middle / bottom, especially useful with rowspan

Don't miss this

Old Attributes You'll Still See in Legacy Code

Attribute	Used for	Use instead
border (on <table>)	Drawing a border	CSS border
cellpadding	Space inside cells	CSS padding
cellspacing	Space between cells	CSS border-spacing
align / valign	Aligning cell content	CSS text-align / vertical-align
bgcolor	Cell/row background color	CSS background
width / height	Sizing cells or table	CSS width / height
summary	Describing table to screen readers	<caption> (removed in HTML5)

Often missed

Accessibility: scope & headers/id

`scope="col"` or `scope="row"` on a `<th>` tells screen readers exactly which cells that header describes — essential once a table has headers on both axes.

`headers` + `id` is the advanced version: each `<td>` lists the ids of the exact header cells it belongs to, for complex multi-level tables.

```
<tr>
  <th scope="col">Name</th>
  <th scope="col">Score</th>
</tr>
<tr>
  <th scope="row">Ali</th>
  <td>88</td>
</tr>
```

WHY IT MATTERS

Sighted users see structure visually. Screen-reader users rely entirely on scope/headers to understand what each number means — this makes your table actually usable for everyone.

Cheat sheet

Every Table Tag & Attribute, In One Place

Tag / Attr	Purpose
<table>	Root container
<caption>	Title (first child)
<thead>	Header row group
<tbody>	Data rows (can repeat)
<tfoot>	Summary row group
<tr>	One table row
<th>	Header cell (semantic)
<td>	Ordinary data cell

Tag / Attr	Purpose
<colgroup>	Groups columns
<col>	One (or span) column(s)
colspan="n"	Merge across n columns
rowspan="n"	Merge across n rows
scope="col row"	A11y: what a <th> labels
headers / id	A11y: explicit cell links
border-collapse	Merge double borders
table-layout: fixed	Predictable col widths

Wrap-up

Recap & In-Class Exercise

You now know:

- `table` / `tr` / `th` / `td` — the structural core
- `thead` / `tbody` / `tfoot` / `caption` — semantic grouping
- `colspan` / `rowspan` — merging cells
- `colgroup` / `col` — column-level styling
- `border-collapse`, `padding`, deprecated attrs, `scope`/headers — styling & accessibility

IN-CLASS EXERCISE

Build a weekly class timetable table: use `<caption>`, `<thead>`/`<tbody>`, at least one `rowspan` for a repeated slot, and style it with CSS (no border attribute!). 15 minutes, then we review together.

Homework: add proper `scope` attributes and a `<tfoot>` summary row to your table.